Technology Stack

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1.	**Frontend / Client-Side** (e.g. React, Angular,HTML/CSS)	Streamlit (UI Components) & Plotly	Enables rapid, responsive web interface development in pure Python with interactive rendering of dynamic retention decay curves and mastery charts.
2.	**Backend / Server-Side** (e.g. Node.js, Python/Django, Java)	Python 3.10+ (NumPy / SciPy)	Powers core backend business logic, mathematical memory decay engines (Ebbinhaus forgetting curves), and asynchronous API handling.

3.	**Database / Data Storage** (e.g. MySQL, MongoDB, PostgreSQL)	SQLite / PostgreSQL	Provides lightweight, reliable relational data mapping for tracking user profiles, historical quiz performances, and retention metrics.
4.	**Cloud / Hosting / Deployment** (e.g. AWS, Azure, Heroku, Docker)	Streamlit Cloud	Offers seamless, zero-downtime hosting directly connected to Git repositories for continuous deployment and easy scalability.
5.	**Version Control & CI/CD** (e.g. GitHub, GitLab, Jenkins)	GitHub & GitHub Actions	Facilitates code versioning, team branch workflows, automated syntax linting, and continuous integration pipelines.
6.	**Third-Party APIs / Other Tools** (e.g. Stripe, Twilio, Google Maps)	Google Gemini API	Drives intelligent, adaptive quiz question generation, contextual hint synthesis, and personalized performance evaluations.